

Dương Bảo Duy

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Summary

Fullstack / AI Software Engineer with 9 years of experience across backend, frontend, mobile, real-time communication, and Linux infrastructure. Strong in Python, TypeScript, Go, Flutter, PostgreSQL, Kubernetes, and self-hosted developer platforms. Recently focused on AI-enabled applications: LLM API routing, agent tooling, document/RAG workflows, speech-to-text experiments, and practical AI integrations that can be shipped as products. Looking for a Fullstack or AI Developer role where I can build production software and turn AI prototypes into reliable user-facing systems.

Work Experience

Freelance Fullstack Developer & AI Builder | Ho Chi Minh City, Vietnam | 12/2022 – Present

- Built and maintained a self-hosted homelab using Linux, QEMU/libvirt, Docker, Kubernetes, NAS storage, and private developer services.
- Developed fullstack prototypes with React/Next.js, Python/FastAPI, Go, PostgreSQL, and containerized deployment workflows.
- Built AI tooling and experiments around LLM routing, local/remote model usage, RAG-style document workflows, speech-to-text, and agent memory.
- Designed developer workflows for personal productivity: GTD knowledge base, local search, file indexing, and agent-assisted coding.
- Practiced production-oriented DevOps: Git-based workflows, service deployment, debugging, observability basics, and Linux server operations.

Senior Software Engineer | Rever Corp | 06/2019 – 12/2022

- Developed and optimized the Rever mobile application with Flutter and backend API integration.
- Improved app performance through refactoring, database query optimization, caching, and practical profiling.
- Implemented automated tests and improved code quality for mobile features used in a production real estate product.
- Integrated REST APIs for data synchronization between mobile clients and backend services.
- Collaborated with product/design teams to deliver user-facing features and resolve production issues.

Senior Software Engineer | VNG Corp | 08/2014 – 05/2019

- Built mobile video call features using WebRTC and native iOS/Android technologies.
- Designed and implemented call quality analysis systems for monitoring real-time communication metrics.
- Developed data processing and backend support tools with Python and Java.
- Wrote C, Java, and Objective-C code to collect, process, and analyze client-side call metrics.

Mobile Developer | BraveSoft Corp | 05/2013 – 07/2014

- Developed Android and iOS applications for client projects.
- Used Android NDK and FFmpeg for low-level media processing features.
- Built an FFmpeg wrapper for video cropping, background insertion, and video merging.

Selected Projects

AI Router / LLM Platform

- Designed a unified layer for working with multiple LLM providers, model routing, API keys, logging, and future cost controls.
- Focus areas: provider abstraction, OpenAI-compatible APIs, request logging, model registry, and practical AI service endpoints.

Agent Tooling / Personal Knowledge System

- Built and operated local workflows for file indexing, GTD task management, notes, memory, and coding-agent assistance.
- Focus areas: SQLite-backed search, Markdown knowledge base, CLI workflows, and agent-readable project context.

Homelab Platform

- Designed and hosted a local Linux/Kubernetes environment for running web apps, backend services, databases, and AI experiments.
- Used QEMU/libvirt, Docker, Kubernetes, NAS storage, Git, Bash, and Linux administration.

Document RAG / AI Workflow Experiments

- Built prototypes for document search, retrieval, summarization, and AI-assisted content workflows.
- Explored OCR, LangChain-style tooling, local knowledge bases, and document automation.

Vietnamese Speech / Whisper Experiments

- Tested speech-to-text workflows for Vietnamese audio and domain-specific voice recognition.

- Explored model adaptation, data collection, and practical STT integration for user-facing apps.

Zalo Call Quality Analysis

- Designed systems to measure and analyze call quality using Python plus native client instrumentation.
- Supported real-time communication reliability for video call features.

Skills

- **Languages:** Python, TypeScript/JavaScript, Go, Java, C/C++, Dart, Kotlin, Swift, Objective-C
- **Frontend:** React, Next.js, Flutter, HTML/CSS
- **Backend:** FastAPI, Django, Go/Gin, Spring Boot, REST APIs, API integration
- **Databases:** PostgreSQL, SQLite
- **AI / ML:** LLM APIs, prompt/context engineering, Whisper/STT, PyTorch, NumPy, Pandas
- **DevOps / Infra:** Linux, Docker, Kubernetes, Bash, self-hosted homelab (QEMU/libvirt/NAS)
- **Domains:** Real-time communication, WebRTC, mobile apps, document automation, agent tooling
- **Languages:** English working proficiency, Vietnamese native, Japanese basic

Education

Bachelor in Mathematics & Computer Science | Ho Chi Minh University of Science | 09/2007 – 05/2013

Certifications

- CCNA, University of Science Computer Center